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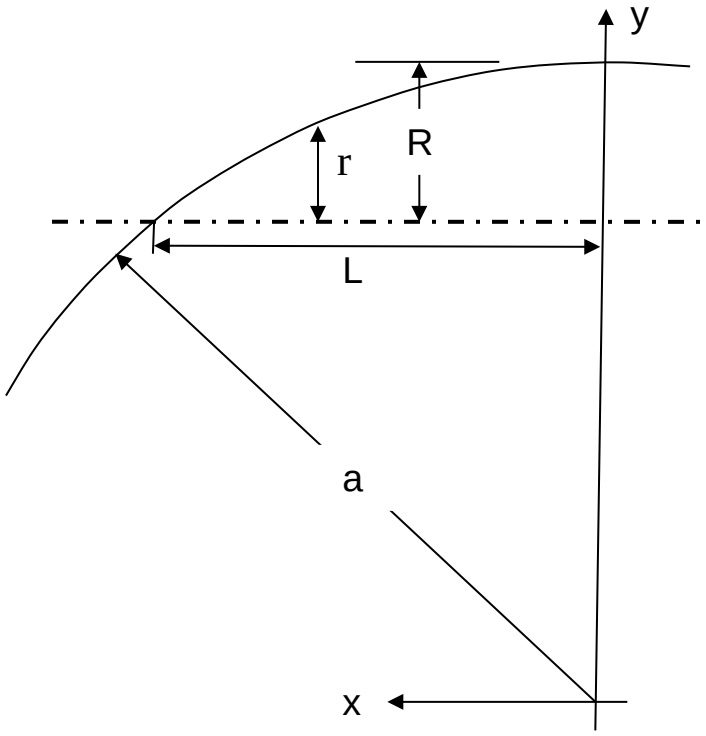
Ogive Nose

06 May 2011

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Nose Profile

The ogive nose form is commonly used in sounding rocket applications. It is formed by a circular arc as sketched below:



The equation for the circular arc is

$$y^2 + x^2 = a^2$$

The relevant condition is

$$y = a - R \text{ when } x = L.$$

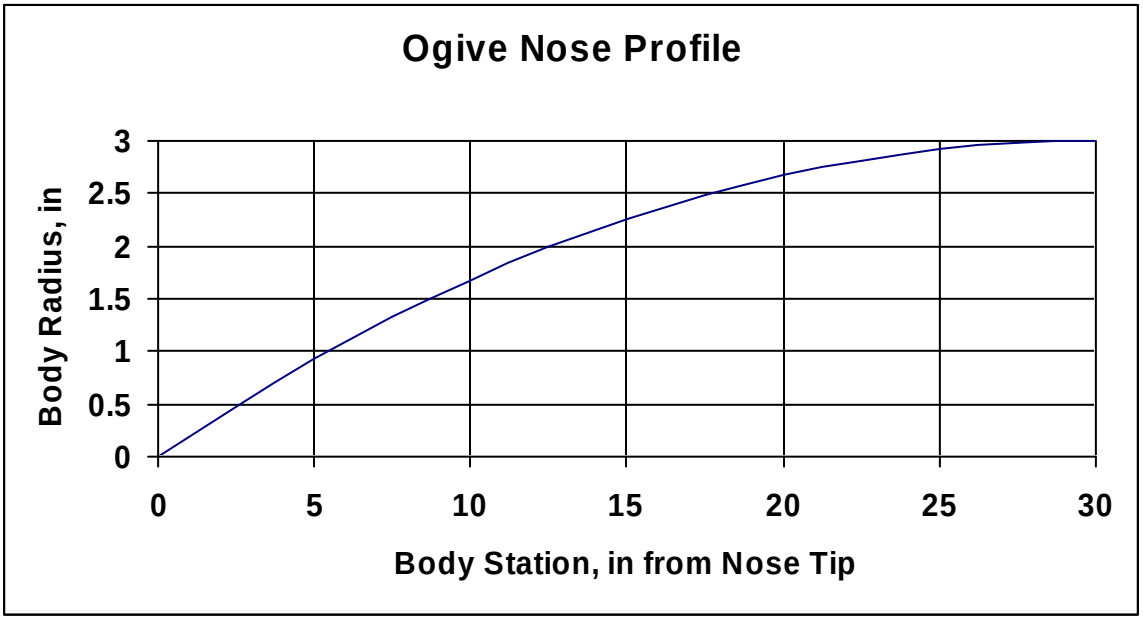
Solving for a gives

$$a = \frac{L^2 + R^2}{2R}.$$

With this, the body radius at x is given by

$$r = R - a + y$$

This equation has been coded into the spread sheet Ogive Profile.xls. A sample sketch for $R = 3$ inches and $L = 30$ inches is shown below.



Surface Area

The ogive surface area is important for two reasons. First, it is the wetted area which drives skin friction drag. Second, when multiplied by wall thickness and material density, it gives the ogive weight. Now, consider a strip of ogive surface of width dx. Its surface area is

$$dS = 2\pi r \frac{\sqrt{dr^2 + dx^2}}{dx} dx, \text{ or}$$

$$dS = 2\pi (R - a + \sqrt{a^2 - x^2}) \sqrt{1 + \frac{x^2}{a^2 - x^2}} dx, \text{ or}$$

$$dS = 2\pi a (1 + \frac{R - a}{\sqrt{a^2 - x^2}}) dx$$

After integrating from $x = 0$ to $x = L$ we find that

$$S = 2\pi a ((R - a) \sin^{-1}(\frac{L}{a}) + L)$$

This result has also been coded into Ogive Profile.xls. For the example sketched above, the surface area is 379.2477 square inches.

Area Centroid

This matters when evaluating the center of mass for an ogive. The moment about the base for an elemental strip is

$$dM = 2\pi ax(1 + \frac{R - a}{\sqrt{a^2 - x^2}})dx$$

After integrating over the ogive length,

$$M = 2\pi a(\frac{1}{2}L^2 + (R - a)(a - \sqrt{a^2 - L^2}))$$

The body station location of the area centroid is

$$\bar{x} = L - \frac{M}{S}$$

This, too, has been coded into the Ogive Profile.xls spread sheet, and, for our example nose, $\bar{x} = 18.70511$ inches aft of the nose tip.